

PART A [25 Marks]

This section consists of **3** questions. Answer **all** questions.

1. (a) Given

$$f(x) = \begin{cases} \frac{x^2 - 9}{x^2 + 2x - 15} & , \quad x \neq 3 \\ C & x = 3 \end{cases}$$

Find the value of C if f is continuous for all x . **[4 marks]**

- (b) The function f is defined by

$$f(x) = \begin{cases} Ax^2 - 6 & , \quad x \leq -\frac{5}{4} \\ Bx + 4 & , \quad -\frac{5}{4} < x < 7 \\ 25 & , \quad x \geq 7 \end{cases}$$

Determine the values of A and B if f is continuous at $x = -\frac{5}{4}$ and $x = 7$.

[5 marks]

2. a) Find the derivative of $f(x) = \frac{1}{x+1}$ by using the first principle.

[4 marks]

- b) Find the derivative of the following functions:

i) $f(x) = \cot \sqrt{4x^2 + 1}$. **[3 marks]**

ii) $f(x) = e^{2x} \ln(3x + 4)$. **[4 marks]**

3. The side of a cube of length x cm is increasing at the rate of 2.5 cm/second. How fast is the volume of the cube increasing when the side is 8 cm long? **[5 marks]**

PART B [75 marks]

This section consists of 7 questions. Answer **all** questions.

1. Given two complex numbers $z_1 = 5 + 3i$ and $z_2 = 2 - i$.
 - a) State $\overline{z_1}$ and $\overline{z_2}$ [1 mark]
 - b) Determine the value of k if $\frac{1}{z_1} = k\overline{z_1}$ [3 marks]
 - c) Find $z_1 z_2$. Hence, show that $\overline{z_1 z_2} = \overline{z_1} \overline{z_2}$ [6 marks]

2.
 - a) Solve the equation $\log(x - 4) + 2\log 3 = 1 + \log\left(\frac{x}{2}\right)$. [5 marks]
 - b) Find the solution set of the inequality $\left|\frac{x - 3}{x + 1}\right| < 2$. [6 marks]

3. Given $h(x) = e^{-2x}$ for domain $(-\infty, \infty)$
 - a) Find $h^{-1}(x)$. State the domain and range [5 marks]
 - b) Sketch the graphs for $h(x)$ and $h^{-1}(x)$. [4 marks]

4. Given $f(x) = \ln(2x + 3)$ and $g(x) = \frac{e^x - 3}{2}$.
 - a) Show that $f(x)$ is one-to-one function algebraically. [3 marks]
 - b) Find $(f \circ g)(x)$ and $(g \circ f)(x)$. Hence, state the conclusion about the results. [6 marks]

5. a) Given
$$f(x) = \begin{cases} \frac{x^3 - 64}{x - 4} & , x \neq 4 \\ 40 & , x = 4 \end{cases}$$
- i) Find $\lim_{x \rightarrow 4} f(x)$. [3 marks]
- ii) Is f continuous at $x = 4$? Give your reason. [3 marks]
- b) Evaluate the following (if exist):
- i)
$$\lim_{x \rightarrow 2} \frac{x^2 + 4x - 12}{|x - 2|}$$
 [5 marks]
- ii)
$$\lim_{x \rightarrow 1} \frac{1 - \sqrt{x}}{1 - x}$$
6. a) Parametric equation of a curve is given by $x = \frac{4t}{t^2 - 1}$ and $y = \frac{3t^2 + 1}{t^2 - 1}$. Find
- (i) $\frac{dy}{dx}$ in terms of t [9 marks]
- (ii) $\frac{d^2y}{dx^2}$ when $t = 2$ [3 marks]
- b) Given $y = (5 - x)e^{-2x}$. Find $\frac{dy}{dx}$ [4 marks]
7. Water is leaking from the bottom of a conical tank with radius 1.5 meter and height 2 meter at a rate of 0.25 cubic meter per minute. The tank was initially full. If the height of water is 1 meter then find the rate of change of
- a) the water level, [7 marks]
- b) the radius of the water surface. [3 marks]

END OF QUESTION PAPER

SUGGESTED ANSWER**PART A**

$$1 \quad \text{a) } C = \frac{3}{4} \quad \text{b) } A = 4; B = 3$$

$$2. \quad \text{a) } f'(x) = -\frac{1}{(x+1)^2}$$

$$\text{b) (i) } f'(x) = -4x \frac{\operatorname{cosec}^2 \sqrt{(4x^2 + 1)}}{\sqrt{(4x^2 + 1)}}$$

$$\text{(ii) } f'(x) = e^{2x} \left(2 \ln(3x + 4) + \frac{3}{(3x + 4)} \right)$$

$$3. \quad \frac{dV}{dt} = 480 \text{ cm}^3 / \text{sec ond}$$

PART B

$$1 \quad \text{a) } \bar{z}_1 = 5 - 3i \quad \bar{z}_2 = 2 + i$$

$$\text{b) } k = \frac{1}{34} \quad \text{c) } z_1 z_2 = 13 - i$$

$$2. \quad \text{a) } x = 9 \quad \text{b) } (-\infty, -5) \cup \left(\frac{1}{3}, \infty \right)$$

$$3. \quad \text{a) } h^{-1}(x) = -\frac{\ln x}{2}$$

$$\text{b) } \text{Domain} = (0, \infty) \quad \text{Range} = (-\infty, \infty)$$

$$4. \quad \text{b) } f \circ g(x) = g \circ f(x) = x$$

5. a) i) $\lim_{x \rightarrow 4} f(x) = 48$ ii) not continuous
- b) i) does not exist ii) $\frac{1}{2}$
6. a) i) $\frac{dy}{dx} = \frac{2t}{(t^2 + 1)}$ ii) $\frac{d^2y}{dx^2} = \frac{27}{250}$
- b) $\frac{dy}{dx} = e^{-2x}(2x - 11)$
7. a) $\frac{dh}{dt} = -\frac{4}{9\pi} m/\text{min}$
- b) $\frac{dr}{dt} = -\frac{1}{3\pi} m/\text{min}$